

GOKUL NEETHIPATHI

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Professional Summary

Senior Data Engineer with 5+ years of experience architecting scalable, cloud-native data platforms on Azure and AWS. Specialized in production-grade ETL/ELT pipelines using Apache Spark, PySpark, Databricks, Snowflake, and Azure Data Factory, with deep expertise in Medallion Architecture, dimensional modeling, and data lakehouse design. Proven track record automating engineering workflows, optimizing query performance, and leading onsite technical initiatives in regulated insurance environments (PII/GDPR/CCPA).

Technical Skills

Data Engineering: ETL/ELT, Apache Spark, PySpark, Databricks, Delta Lake, Medallion Architecture (Bronze/Silver/Gold), dbt, Hadoop, Hive, Spark Structured Streaming, Apache Kafka, Azure Event Hubs.

Data Warehousing & Modeling: Snowflake, Snowpark, Dimensional Modeling, Star/Snowflake Schema, Fact/Dimension Tables, SCD Type 1/2, Data Vault, OLAP/OLTP, Data Lakehouse.

Orchestration: Azure Data Factory (ADF), Apache Airflow, Azure Logic Apps, AWS Step Functions.

Cloud (Azure): ADF, Azure Databricks, ADLS Gen2, Azure Blob Storage, Event Hubs, Logic Apps, Synapse Analytics, Key Vault, Unity Catalog.

Cloud (AWS): S3, Lambda, DynamoDB, Step Functions, CloudWatch.

Programming: Python (Pandas, NumPy, PySpark), Advanced SQL (CTEs, Window Functions, Query Optimization, Stored Procedures), Scala, Java, JavaScript.

File Formats: Parquet, Avro, ORC, Delta, JSON, CSV.

DevOps & CI/CD: Git, GitHub Actions, Azure DevOps, Jenkins, Docker, Kubernetes, Terraform, Agile/Scrum, SDLC.

Data Governance & Quality: Unity Catalog, Great Expectations, Data Lineage, PII/GDPR/CCPA Compliance.

Databases & BI: Snowflake, MongoDB, DynamoDB, Firebase | Power BI, Streamlit, MS Excel.

GenAI (Applied to Data Pipelines): RAG, LangChain, LangGraph, Agentic AI, TensorFlow Lite.

Professional Experience

INFOSYS LIMITED, Raleigh, NC & Chennai, India

Jan 2021 – Present

Senior Data Engineer (Specialist Programmer)

Client: Arch Insurance (Raleigh, NC / Chennai)

Jun 2022 – Present

- **Lead Data Engineer:** Primary onsite technical liaison translating complex underwriting and actuarial requirements into scalable lakehouse architectures on Azure Databricks and Snowflake.
- **Data Architecture (Medallion):** Designed and deployed enterprise ETL/ELT pipelines using Azure Data Factory, Databricks, and PySpark, processing 2TB+ of structured and semi-structured insurance data daily across Bronze, Silver, and Gold layers on ADLS Gen2 in Parquet/Delta formats.
- **Dimensional Modeling:** Built Star Schema models with SCD Type 2 on Snowflake to support underwriting analytics, enabling sub-second query response times for downstream Power BI dashboards.
- **Performance Tuning:** Optimized Spark and Snowflake workloads via Z-ordering, partitioning, clustering keys, broadcast joins, and Adaptive Query Execution (AQE), reducing query latency by approximately 40% and cutting Snowflake compute credits by approximately 25%.
- **Automation Tooling:** Built a custom Deployment Management Portal (Python, Streamlit, Flask) integrating Jira, Egnyte, and Git APIs to automate the monthly release lifecycle, cutting deployment coordination time by 90% and eliminating manual errors.
- **Framework Development:** Built a reusable, metadata-driven PySpark ingestion framework with built-in data quality checks (Great Expectations) and lineage tracking, standardizing team practices and reducing new pipeline development time by approximately 60%.
- **CI/CD:** Implemented CI/CD pipelines using Azure DevOps and Git for Databricks notebooks and ADF artifacts, enabling automated testing and zero-downtime deployments across Dev/QA/Prod.
- **Cloud Migration:** Spearheaded migration of legacy on-premise Hive datasets to Snowflake using Snowpark, applying advanced clustering and partitioning strategies to lower TCO with 100% data parity.
- **Data Governance:** Enforced PII/GDPR/CCPA compliance via Unity Catalog with fine-grained access control, column-level masking, and end-to-end data lineage in a regulated insurance environment.

Client: State Farm Insurance (Chennai)

Dec 2021 – May 2022

- **Infrastructure as Code:** Provisioned scalable AWS infrastructure using Terraform, ensuring consistent, version-controlled environments across Dev, QA, and Production.

- **Serverless Data APIs:** Developed Python-based AWS Lambda APIs integrated with DynamoDB and S3 to handle Data Rights Management (DSAR) requests, enabling a fully serverless, event-driven architecture aligned with CCPA/GDPR.
- **Workflow Orchestration:** Authored AWS Step Functions and CloudWatch-driven scheduling for downstream data jobs, ensuring SLA-compliant execution.

Client: Allstate Insurance (Chennai)

Jan 2021 – Nov 2021

- **Storage Migration (HDFS to Scality):** Executed a high-stakes Hadoop-to-Scality migration of multi-TB datasets in Parquet/ORC formats, ensuring 100% data integrity through automated reconciliation.
- **Legacy Modernization:** Replaced manual underwriting validation with automated Python and PySpark scripts, increasing data-quality coverage and freeing analysts from repetitive QA work.
- **Data Purging:** Wrote optimized Scala/Spark scripts to identify and purge obsolete MongoDB records, significantly reducing storage overhead prior to migration.
- **Data Validation:** Built Python/SQL validation frameworks to reconcile legacy Hive tables against Scality datasets, ensuring 100% post-migration accuracy.

Education

Bachelor of Engineering (B.E.)

2016 – 2020

Sri Ramakrishna Engineering College, Coimbatore

- **Awards:** Recipient of the Distinguished Alumni Award (2020) for overall academic and extracurricular excellence.

Earlier Experience

STATE OF TAMIL NADU, INDIA

Apr 2020 – Jul 2020

Software Developer Intern

- Developed a native Android application using Java and Android Studio to monitor quarantine compliance during the COVID-19 pandemic, working directly with government officials to define feature requirements.

INOVARK TECHNOLOGIES

Jan 2018 – Oct 2019

Software Developer Intern

- Engineered a TensorFlow-based object detection system to identify trespassing and automatically alert security personnel, enhancing campus safety.

Leadership

- **Technical Interviewer & Mentor, Infosys Limited (2021–Present):** Conduct technical interviews for lateral and fresh talent; onboard and mentor new recruits on complex insurance data engineering projects.
- **Project Lead, Brain Computer Interface (UG Capstone) (2019–2020):** Led end-to-end execution of the BCI research project; responsible for task allocation, timeline management, and delivering technical presentations to college and conference boards (resulting in two IEEE publications).
- **Placement Coordinator, Sri Ramakrishna Engineering College (2016–2020):** Facilitated campus recruitment drives and mentored 100+ peers on technical interview preparation, coding standards, and resume building.
- **Executive Member, Computer Society of India (2017–2019):** Organized technical workshops and seminars for a student body of 500+ members, fostering a culture of technical excellence and peer learning.

Publications & Research

Publications

- **"Real-Time Thought Controlled Action for Paralytic Patient using EEG"** – 2nd International Conference for Emerging Technology (INCET), 2021. Brain-Computer Interface enabling paralyzed patients to control gaming interfaces in real time.
- **"Real-Time Electric Light Control using EOG Signals"** – 2nd International Conference for Emerging Technology (INCET), 2021. EOG-based system for hands-free home automation aimed at users with mobility impairments.

Key Academic Projects

- **Perimeter Fencing (Computer Vision):** Engineered a TensorFlow-based object detection system to identify trespassing and automatically alert security personnel, enhancing campus safety.
- **Vehicle Tracking Application:** Deployed a real-time Android app using GPS and Google Maps API for the university fleet, enabling students to track live bus locations and administrators to monitor route adherence.
- **Accident Monitoring System (IoT):** Developed an IoT system to detect vehicle accidents and dynamically manipulate traffic signals based on ambulance location to reduce emergency response times.